

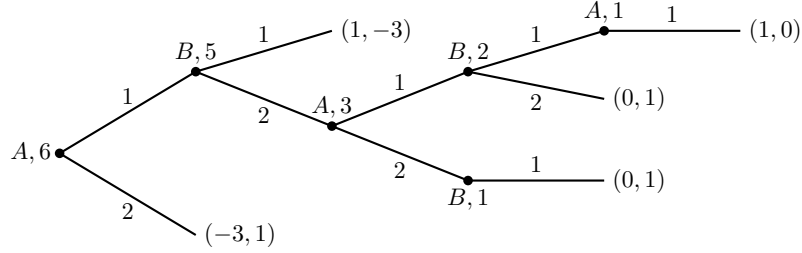
## Nash equilibria and subgame-perfect Nash equilibria

Suppose that two players,  $A$  and  $B$ , play the following game. There are 6 juice bottles in the house.

- Player  $A$  moves first
  - Each player can take either 1 or 2 juice bottles on his turn
  - After each move, the other player observes what happened and then takes his turn
  - If, after a player moves, exactly 4 juice bottles remain, the game ends. The player who caused the stock to reach 4 has to go buy more juice and receives payoff  $-3$ . The other player receives payoff 1
  - In every other case, the player who takes the last juice bottle receives payoff 1, and the other player receives payoff 0
1. Write the game in extensive form
  2. Write the actions and strategies of both players
  3. Find the Nash equilibria
  4. Find the subgame-perfect Nash equilibria
  5. Comment on whether your answers in c) and d) are consistent in terms of the number of strategies found

## Solution

1. The game in extensive form is:



2. Player  $A$  moves at the initial node with 6 bottles left, and may also move later with 3 bottles left. If only 1 bottle remains,  $A$  has only one possible action. Therefore, we write  $A$ 's strategies as

$$s_A = (a_6, a_3)$$

where  $a_6 \in \{1, 2\}$  and  $a_3 \in \{1, 2\}$ . Thus,

$$S_A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$$

Player  $B$  may move with 5 bottles left and with 2 bottles left. If only 1 bottle remains,  $B$  has only one possible action. Therefore, we write  $B$ 's strategies as

$$s_B = (b_5, b_2)$$

where  $b_5 \in \{1, 2\}$  and  $b_2 \in \{1, 2\}$ . Thus,

$$S_B = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$$

**Conclusion:** each strategy is a complete contingent plan, specifying what the player would do at every decision node

3. The strategic-form game is:

|        | (1, 1)           | (1, 2)           | (2, 1)          | (2, 2)                  |
|--------|------------------|------------------|-----------------|-------------------------|
| (1, 1) | ( <u>1</u> , -3) | ( <u>1</u> , -3) | ( <u>1</u> , 0) | ( <u>0</u> , <u>1</u> ) |
| (1, 2) | ( <u>1</u> , -3) | ( <u>1</u> , -3) | (0, <u>1</u> )  | ( <u>0</u> , <u>1</u> ) |
| (2, 1) | (-3, <u>1</u> )  | (-3, <u>1</u> )  | (-3, <u>1</u> ) | (-3, <u>1</u> )         |
| (2, 2) | (-3, <u>1</u> )  | (-3, <u>1</u> )  | (-3, <u>1</u> ) | (-3, <u>1</u> )         |

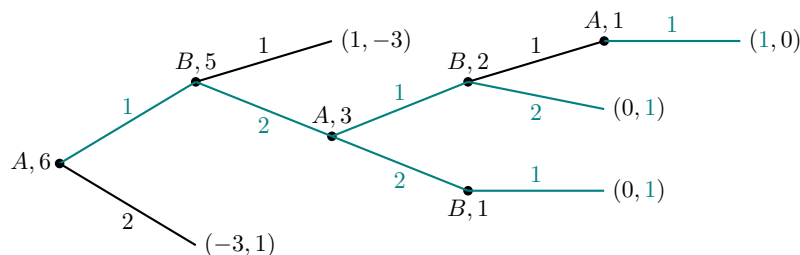
The cells in which both payoffs are underlined are the Nash equilibria. Therefore, the Nash equilibria are

$$((1, 1), (2, 2)) \quad ((1, 2), (2, 2))$$

**Conclusion:** the game has two Nash equilibria in pure strategies:

$$((1, 1), (2, 2)) \quad ((1, 2), (2, 2))$$

4. We solve the game by backward induction.



If  $A$  moves with 1 bottle left,  $A$  takes it and obtains

$$(1, 0)$$

If  $B$  moves with 1 bottle left,  $B$  takes it and obtains

$$(0, 1)$$

Now consider  $B$ 's decision when 2 bottles remain. If  $B$  takes 1, then  $A$  takes the last bottle, so the payoff is

$$(1, 0)$$

If  $B$  takes 2,  $B$  takes the last bottle, so the payoff is

$$(0, 1)$$

Thus,  $B$  chooses 2.

Now consider  $A$ 's decision when 3 bottles remain. If  $A$  takes 1, then  $B$  takes 2 and the payoff is

$$(0, 1)$$

If  $A$  takes 2, then  $B$  takes the last bottle and the payoff is also

$$(0, 1)$$

Thus,  $A$  is indifferent between taking 1 and taking 2.

Now consider  $B$ 's decision when 5 bottles remain. If  $B$  takes 1, exactly 4 bottles remain, so  $B$  receives  $-3$ . If  $B$  takes 2, the continuation payoff gives  $B$  payoff 1. Hence,

$$1 > -3$$

and  $B$  chooses 2.

Finally, at the initial node, if  $A$  takes 2, exactly 4 bottles remain, so  $A$  receives  $-3$ . If  $A$  takes 1, then  $B$  takes 2, and  $A$  receives 0. Hence,

$$0 > -3$$

and  $A$  chooses 1.

Therefore, the subgame-perfect Nash equilibria are

$$((1, 1), (2, 2)) \quad ((1, 2), (2, 2))$$

**Conclusion:** the game has two subgame-perfect Nash equilibria:

$$((1, 1), (2, 2)) \quad ((1, 2), (2, 2))$$

The multiplicity comes from  $A$ 's indifference when 3 bottles remain

5. The answers in parts c) and d) are consistent. Both parts give the same two strategy profiles:

$$((1, 1), (2, 2)) \quad ((1, 2), (2, 2))$$

This is possible because every Nash equilibrium found in the strategic form also prescribes optimal behavior in every subgame.

**Conclusion:** in this game, the Nash equilibria and the subgame-perfect Nash equilibria coincide. Therefore, the number of equilibria is the same in c) and d)